1.0 Introduction

- 1.1 External Motivation
- 1.2 Related Work
- 1.3 Review Methodology
- 1.4 Document Structure

2.0 Visual Inspections

2.2 Inspected Assets

2.1 Electrical Systems

2.3 Process Automating

3.0 Data Acquisition

- 3.1 Technologies & Sensors
- 3.2 Means & Platforms
- 3.3 Scale & Distance

4.0 Data Processing

- 4.1 Deep Vision Evolution
- 4.2 Algorithms & Architectures
- 4.3 Emerging Methodologies

5.0 Applications Scenarios

5.1 Power Generation

- 5.1.1 PhotoVoltaic Modules
- 5.1.2 Wind Turbine Blades
- 5.1.3 Hydroelectric Dam Walls

5.2 Transmission & Distribution

- 5.2.1 Power Lines & Structures
- 5.2.2 HV Insulators & Components
- 5.2.3 Substations & Transformers

6.0 Current Challenges

- 6.1 Operational Viability
- 6.2 Data Quality Measurement
- 6.3 Performance Evaluation

7.0 Future Directions

- 7.1 Embedded Vision
- 7.2 Multimodal Data
- 7.3 Active Learning

8 Conclusion